

Rushil Singh

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Information Systems, Minor in Artificial Intelligence

Expected May 2028

Relevant Coursework: Data Structures & Algorithms, Database Design & Development, Natural Language Processing, Designing Human-Centered Software, Discrete Math, Probability Theory, Linear Algebra, Calculus 3

PROFESSIONAL EXPERIENCE

Twinly

Remote, EST

Software Engineer

June 2026 – Present

- Designed and shipped cross-platform AI agent capabilities in Swift and Python including voice cloning, reservation booking, and automated gaming accessible through natural prompts, supporting 100+ users and 2,000+ followers
- Deployed Electron desktop builds across macOS and Windows, debugging platform-specific issues and maintaining consistently updated releases through Git version control, code reviews, and PR/MR-based feature integration

Perforated AI

Pittsburgh, PA

Machine Learning Research Collaborator

May 2026 – Present

- Developed PyTorch-based transformer optimization techniques using Perforated Backpropagation and artificial dendrite architectures, reducing model parameters by 60% while accelerating trading-model inference by 15%+
- Designed W&B hyperparameter sweeps and multi-seed experimentation across 15+ parameter combinations to optimize dendrite thresholds, regularization, and module placement against baseline performance

ListenFirst

New York City, NY

AI Engineering Intern

July 2026 – August 2026

- Built AI-powered workflows shipped as organization-wide Claude Skills through Python, JavaScript, and RESTful APIs that reduced manual reporting effort by 75% across 60+ projects as the Partnerships unit's sole intern
- Developed production automation for monthly billing and scheduling across 40+ client accounts, cutting API call volume by 90%+ through optimized data pipelines and surfacing \$15K+ in billing discrepancies missed manually

CAMPUS INVOLVEMENT

180 Degrees Consulting

Pittsburgh, PA

Student Consultant

Active Member

- Selected from a competitive applicant pool to undergo new consultant training involving developing public speaking skills, learning deck design, and more in preparation to consult non-profits in Pittsburgh on real-world projects

Traders @ CMU

Pittsburgh, PA

Prediction Market Analyst

Active Member

- Built a cryptocurrency arbitrage tool achieving a 65% win rate through fine-tuning an ML model and backtesting with trade simulations to optimize discrepancies between Kalshi's BTC price prediction market and its sourcing

PROJECTS

TuneBloom – Music Discovery Application | *JavaScript, React, Tailwind CSS, RESTful API, PostgreSQL* | Demo Video

- Engineered a content-based personalized music recommendation engine using Euclidean distance algorithms on Spotify audio features, with fallback strategies and exclusion logic to filter through 1,000+ previously-heard tracks
- Refined a gamified UI/UX system with Framer Motion and real-time feedback loops to make recommendations behind secure login authentication and an efficient API integration layer with smart caching and batch processing

Option(al) Risk – Quantitative Finance Tool | *TypeScript, Python, CSS, Flask, React, RESTful API* | Live Demo

- Created a full-stack web dashboard to deliver options portfolio risk analytics on data-backed volatility calculations, performing the Black-Scholes model across 61+ stock tickers to allow traders to analyze portfolio sensitivity
- Implemented a Monte Carlo risk simulation engine processing 10,000+ price path scenarios to calculate Value-at-Risk statistics at 95th and 99th percentiles, providing downside risk assessment for options strategies

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, Swift, SQL, HTML, CSS

Frameworks & Libraries: PyTorch, React, Node.js, Flask, Pandas, NumPy, Matplotlib, Framer Motion, Tailwind CSS

Developer Tools: Git, GitHub, Linux, Docker, GitHub Actions, Supabase, PostgreSQL, Jupyter, Claude Code